

കേരളപ്പിറവി ആശംസകൾ

Kerala Piravi ashamsakal

1 Nov 1956 → 1 Nov 2026 · Kerala turns 70

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AI · From Hype to BLDing

your first voice agents

Abhishek Mishra · @stalwartcoder

DevRel & DevEx · Smallest AI · BUILD Day Cochin

MEET CHECHI

+91 22 6809 3636

Before we build anything, let's call her.

TODAY

Five things, in this order

1. **Meet Chechi** · a quick call · ~1 min → phone
2. **Build** your first voice agent · ~12 min → laptops out
3. **Look underneath**, with open source · ~12 min
4. **Demo and test** · Chechi again, then pytest · ~7 min
5. **Your turn** · take the repo home

STEP 0 · GET IN

Scan, sign up, you’re in



app.smallest.ai

Any email works

Free to start, no card

Slow wifi? One phone hotspot for three of you

BUILD ALONG · FOLLOW MY SCREEN

Build Kochi Buddy

1 · Create Agent (green, top right) → Start from Scratch → Single Prompt

2 · Name it `kochi-buddy` · Call Direction Inbound

3 · Pick a voice, hit preview · Malayalam (Beta) is in Languages, try it

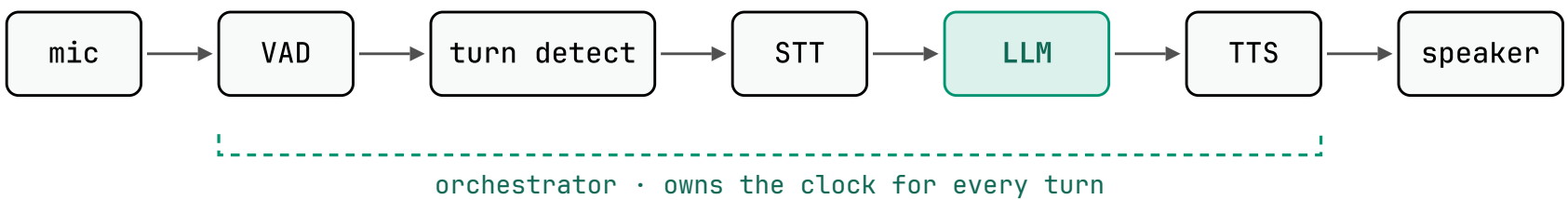
4 · Paste the prompt from the repo · add a first message · Save

5 · Test Agent → Web Call → allow the mic → talk

> Try three things. A normal question. A weird one ("book me a flight to Mars"). Then interrupt it mid sentence.

THE BRIDGE

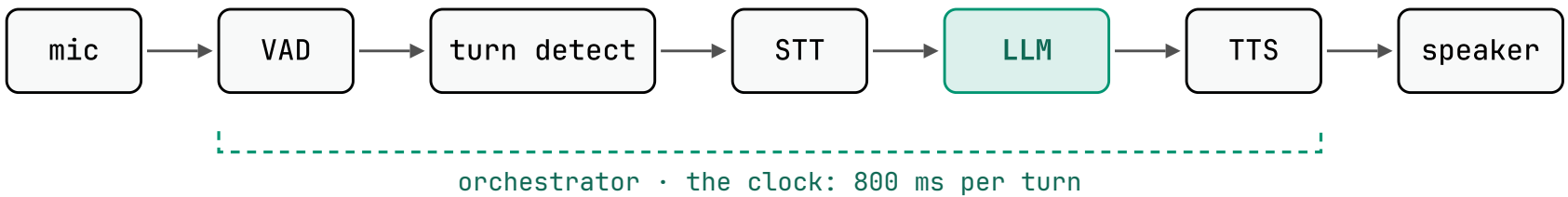
You just used a platform. Here's what's underneath.



THE MENTAL MODEL EVERYONE STARTS WITH



THE MENTAL MODEL YOU ACTUALLY NEED



Everything hard about voice AI lives in the arrows, not the boxes.

~200ms

average gap between human turns, 10 languages

Stivers et al., PNAS 2009

1,500ms

"an important target to aim for", voice to voice

Voice AI & Voice Agents primer

?ms

what your agent actually does

you'll feel it on the phone demo

Everything must stream

✗ `record → transcribe → think → synthesize → play`

✓ `stream → stream → stream → stream → stream`

Every component emits partial results. Nothing waits for "done".

mic | VAD | turn | STT | LLM | TTS | spk

STOP 1 · TRANSPORT

How your voice reaches the code

	Use for	Why
WebRTC	Browser, mobile	Opus codec, jitter buffer, packet-loss concealment, echo cancellation
WebSocket	Server to server, telephony bridges	Simple, no media features built in
SIP / PSTN	Phone calls	How the phone network works. Narrowband audio (8 kHz).

> Do not build WebRTC yourself. NAT traversal, STUN/TURN, ICE, codec negotiation.

mic | VAD | turn | **STT** | LLM | TTS | spk

STOP 2 · VAD, THEN SPEECH TO TEXT

Your sentence becomes words, while you speak

VAD decides "is someone speaking right now?" It gates everything downstream.

Streaming STT emits partial transcripts as audio arrives. Not after you stop.

> Batch-first models (Whisper-style) want the whole utterance. A live conversation can't give them that.

Open source · Silero VAD · Whisper · AI4Bharat IndicConformer

mic VAD turn STT **LLM** TTS spk

STOP 3 · THE LLM, PROMPTED FOR THE EAR

The reply must be speakable

Text agent	Voice agent
Bullet points, markdown, emoji	Never
"₹1,234.50"	"twelve hundred thirty-four rupees and fifty paise"
Long, thorough answers	1 or 2 sentences, then stop
"Let me know if you need anything else!"	Once. Not every turn.

mic VAD turn STT LLM **TTS** spk

STOP 4 · TEXT TO SPEECH, THEN PLAYBACK

The reply becomes a voice

TTS starts on the **first chunk** of LLM output, not the full reply.

Audio streams back over the same transport and plays while the rest is still being generated.

> The metric that matters is **TTFB**, time to first byte. Not total synthesis time.

Open source · Piper · Kokoro · IndicF5



THE WHOLE AGENT AS CODE · PIPECAT

One pipeline, interchangeable models

```
stt = SmallestSTTService(api_key=SMALLEST_KEY)                    # Pulse
llm = AnthropicLLMService(settings=AnthropicLLMService.Settings(model="claude-haiku-4-5-20251001"))
tts = SmallestTTSService(api_key=SMALLEST_KEY, settings=SmallestTTSSettings(voice="meher"))

# the swap: one line, pipeline untouched
# tts = SmallestTTSService(api_key=SMALLEST_KEY, settings=SmallestTTSSettings(voice="nolan"))
# llm = OpenAILLMService(...)   tts = CartesiaTTSService(...)   stt = DeepgramSTTService(...)

pipeline = Pipeline([transport.input(), stt, ctx.user(), llm, tts,
                     transport.output(), ctx.assistant()])
```

The orchestrator owns transport, turn-taking, interruption and context. **Pipecat** here. **LiveKit Agents** is the other big one. Both are open source.

HARD PART 1 · KNOWING WHEN THE HUMAN STOPPED TALKING

"I'd like a large pizza…"

…done? Or

"I'd like a large pizza… and garlic bread."

Silence timer → semantic endpointing → semantic + acoustic. `Pipecat Smart Turn · LiveKit turn detector`

HARD PART 2 · BARGE-IN, AND THE CONTEXT BUG

What did the agent actually say?

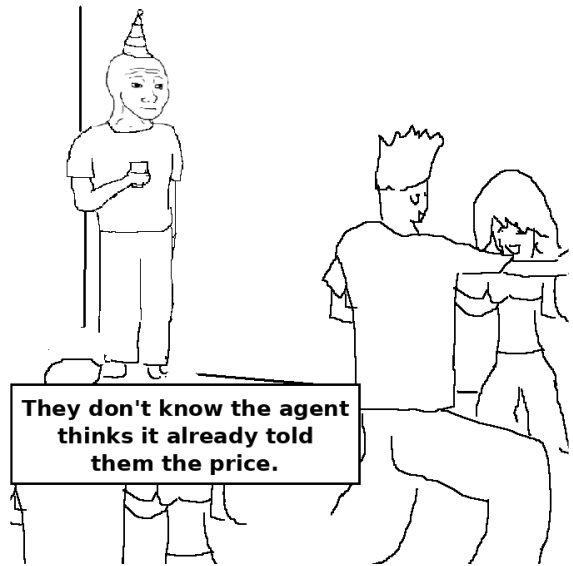
Agent: *"Sure, I can book that for you on Tuesday at 3pm, and I'll send a confirm..."*

User cuts in at **"Tuesday."**

What does the conversation history say the agent said?

✕ The full sentence. The agent now believes 3pm is agreed. It is not.

✓ Truncate to what was **actually played out loud**.



THE SHOPPING LIST

Ten building blocks

1 · Transport
WebRTC / WS / SIP. How audio moves.

2 · VAD + turn detection
When to listen, when to speak.

3 · STT
The ears. Streaming, always.

4 · LLM + voice prompt
The brain, prompted for the ear.

5 · TTS
The voice. TTFB is the metric.

6 · Orchestrator
The conductor. Pipecat or LiveKit Agents.

7 · Tools / APIs
The hands. Async, masked.

8 · Knowledge / RAG
The memory. Keep retrieval tight.

9 · Telephony
Optional. A number and a carrier.

10 · Evals + observability
Latency percentiles. Listen to calls.

The framework hands you 1, 2 and 6. You choose 3, 4, 5. You build 7 and 8. You rent 9. You never stop doing 10.

LIVE · CHECHI SPEAKS MANGLISH

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I call, you shout the questions. English, Hindi or Malayalam.

It's an AI. Normal call charges. Try it yourself after the session, one call at a time.

TEST IT

How do you know it still works tomorrow?

```python
def test_brain_replies_in_manglish_and_keeps_it_short():
    reply = chechi.think("ചേച്ചി, ഇന്ന് Kerala Piravi ആണല്ലോ. ഒരു hi പറയാമോ?")
    assert MALAYALAM.search(reply)          # Malayalam in the reply
    assert len(reply) < 400                 # short enough to say out loud

@pytest.mark.live
def test_live_chechi_answers_the_phone():
    result = sipling.probe(uri, transport="tcp", timeout=15)   # a real SIP call
```

offline: `pytest -v test_chechi_call.py`

phone: `pytest -v --live -k live test_chechi_call.py`

sipling · open source · github.com/abhishekmishragithub/sipling

WHERE TO ACTUALLY LEARN THIS

Resources

The textbook: *Voice AI & Voice Agents: An Illustrated Primer* · `voiceaiandvoiceagents.com`

Pipecat: `docs.pipecat.ai` · `pip install "pipecat-ai[smallest]"`

LiveKit Agents: `docs.livekit.io/agents` · `pip install livekit-plugins-smallestai`

Smallest Waves docs: `docs.smallest.ai`

the primer

pipecat

livekit agents

smallest docs

TAKE IT HOME

Everything from today is in here



1 · Finish your agent at home

2 · Run Chechi from the repo

3 · Post it, tag `@stalwartcoder` `#BUILDDayCochin`

`github.com/abhishekmishragithub/workshops · kochi-build-day-2026`



Questions

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@stalwartcoder
DevRel & DevEx · Smallest AI

Nanni, Kochi! Kerala Piravi ashamsakal